

Temperature-Dependent Bulk Modulus as the Origin of Dark Energy and Dark Matter

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Working Paper in the Dynamic Aether Mechanics Framework

Abstract

Paper 6 identifies dark energy as the thermal pressure of hot void Aether and dark matter as the two-fluid first-sound correction from an outward temperature gradient. This working paper explores an alternative thermal landscape in which matter-containing regions are *warmer* than voids—heated by viscous dissipation of gravitational inflow—while voids cool adiabatically through expansion. If the Aether bulk modulus K_A increases with decreasing temperature (as the first-sound speed does in superfluid helium-4), the wave speed $c = \sqrt{K_A/\rho_A}$ is higher in cold voids than near warm mass. On galactic scales, the outward-increasing c produces dark matter through the acoustic metric ($\Delta K_A/K_A \sim 4 \times 10^{-6}$). On cosmic scales, the growing K_A contrast from a cooling cascade drives accelerating expansion ($\Delta K_A/K_A \sim 2 \times 10^{-4}$). A single parameter $\eta_K \sim 10^{-4} \text{ K}^{-1}$ —matching the measured temperature sensitivity of helium-4 at low temperature—governs both phenomena. The cascade naturally produces the phantom-to-quintessence crossing observed by DESI. The local Aether temperature at the Sun’s position ($\sim 2.75 \text{ K}$) matches T_{CMB} to within 1%.

This working paper explores an alternative temperature model within the Dynamic Aether Mechanics framework. It references results from Paper 1 (General Overview), Paper 3 (Gravity), Paper 6 (Cosmology), and Paper 6(a) (Cooling Feedback Model) of the main series. The ideas presented here are under development and may be incorporated into the main papers in revised form.

1. Introduction

Paper 6 establishes that the Aether possesses three independently varying properties—density, pressure, and temperature—decoupled by the mechanism of permanent binding [5]. In that framework, matter-containing regions are *cold* (thermal energy locked into mass by binding) and cosmic voids are *hot* (heated by stellar radiation and bulk-viscous dissipation of the Hubble flow). This temperature landscape produces dark energy (thermal pressure in hot voids driving expansion) and dark matter (the two-fluid first-sound correction from the outward temperature gradient) [7].

This paper asks: *what if the temperature gradient is reversed?*

Three observations motivate the question:

1. **Matter creation was early.** Permanent binding occurred primarily in the first few billion years of cosmic history. The thermal deficit it created was enormous but finite, and stellar radiation plus viscous dissipation have been returning energy to the near-mass Aether for over ten billion years. The initial cooling from binding may be long since compensated.
2. **Gravitational inflow creates large velocity gradients.** Near a Milky-Way-sized galaxy, the inflow velocity gradient $dv/dr \sim v/r \sim 200 \text{ km/s}/10 \text{ kpc} \sim 6 \times 10^{-16} \text{ s}^{-1}$ is approximately 100 times larger than the Hubble flow gradient $\nabla \cdot \mathbf{v} = 3H \sim 7 \times 10^{-18} \text{ s}^{-1}$. Viscous dissipation is proportional to the square of the velocity gradient, so the heating rate near mass can dominate over the heating rate in voids by a factor of $\sim 10^4$.
3. **Superfluid helium-4 is stiffer when cold.** The first-sound speed in He-4 is higher at lower temperatures: cold superfluid has a larger effective bulk modulus. If the Aether shares this property, cold voids would have higher K_A and higher wave speed c than warm matter-containing regions.

The resulting model—warm near mass, cold in voids, with temperature-dependent K_A —produces both dark matter and dark energy from a single mechanism, governed by one parameter. The remainder of this paper develops the quantitative framework and checks its consistency with observations.

2. The Hot Matter Model

2.1 Temperature Reversal

In the standard Paper 6 model, permanent binding continuously removes thermal energy from the Aether near matter, keeping mass-containing regions cold. The alternative proposed here is that this thermal deficit has been more than compensated by viscous heating from gravitational flows.

The normal (viscous) component of the two-fluid Aether dissipates kinetic energy from any flow with velocity gradients. The dissipation rate per unit volume is:

$$\dot{\varepsilon} = \zeta_n (\nabla \cdot \mathbf{v})^2 + \mu_n \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3} \delta_{ij} \nabla \cdot \mathbf{v} \right)^2 \quad (1)$$

Near galaxies, gravitational inflow produces both shear and compressional velocity gradients of order $v/r \sim 6 \times 10^{-16} \text{ s}^{-1}$ (for a Milky-Way-sized galaxy at 10 kpc). In cosmic voids, the Hubble

flow is pure dilation with $\nabla \cdot \mathbf{v} = 3H \sim 7 \times 10^{-18} \text{ s}^{-1}$ and zero shear. The ratio of dissipation rates scales as the square of the velocity gradients:

$$\frac{\dot{\epsilon}_{\text{mass}}}{\dot{\epsilon}_{\text{void}}} \sim \left(\frac{6 \times 10^{-16}}{7 \times 10^{-18}} \right)^2 \sim 10^4 \quad (2)$$

Even after accounting for the lower normal fraction near mass (which reduces ζ_n and μ_n), the gravitational flow gradients are so much larger that the near-mass heating rate can dominate.

2.2 Self-Reinforcing Feedback

The reversed temperature landscape is maintained by a positive feedback loop on both sides:

1. **Near mass (warm).** Higher temperature $\Rightarrow$ larger normal fraction $\rho_n/\rho_A \Rightarrow$ larger bulk and shear viscosity $\Rightarrow$ more dissipation from gravitational inflow $\Rightarrow$ more heating $\Rightarrow$ stays warm.
2. **In voids (cold).** Lower temperature $\Rightarrow$ smaller normal fraction $\Rightarrow$ lower viscosity $\Rightarrow$ negligible dissipation (only the weak Hubble-flow dilation contributes, and even that is suppressed by the low ρ_n) $\Rightarrow$ expansion cools the void further $\Rightarrow$ stays cold.

This is a stable configuration: neither side has a mechanism to reverse its thermal state. The warm region cannot cool (heating exceeds any losses), and the cold region cannot warm (no significant heat source). By contrast, the standard Paper 6 model must explain how voids remain hot despite adiabatic cooling from expansion—a question addressed by Paper 6(a)’s roton freeze-out mechanism [8].

2.3 Temperature-Dependent Bulk Modulus

In superfluid helium-4, the first-sound speed $c_1 = \sqrt{K/\rho}$ is higher at lower temperatures. At $T = 0$, the superfluid is maximally stiff; as temperature increases toward the lambda point ($T_\lambda = 2.17 \text{ K}$), thermal excitations soften the medium and c_1 decreases. The fractional sensitivity at low temperature ($T < 1 \text{ K}$) is approximately $(1/c_1^2)|dc_1^2/dT| \sim 10^{-4} \text{ K}^{-1}$.

If the Aether exhibits the same behavior, the bulk modulus depends on temperature:

$$K_A(T) = K_0 [1 - \eta_K (T - T_{\text{ref}})] \quad (3)$$

where $K_0 \sim 10^{28} \text{ J/m}^3$ is the bulk modulus at the reference temperature, and

$$\eta_K \equiv \frac{1}{K_A} \left| \frac{dK_A}{dT} \right| \quad (4)$$

is the fractional temperature sensitivity (units: K^{-1}). The wave speed becomes position-dependent through the temperature field:

$$c(r)^2 = \frac{K_A(T(r))}{\rho_A} \quad (5)$$

In the reversed model: near mass (warm, higher T) $\Rightarrow$ lower $K_A \Rightarrow$ lower c . In voids (cold, lower T) $\Rightarrow$ higher $K_A \Rightarrow$ higher c . The wave speed *increases outward* from galaxies—precisely the gradient needed for the dark matter effect in the acoustic metric.

The equation of state becomes $P = K_A(T) \ln(\rho/\rho_0)$, coupling temperature to mechanical pressure. This is a departure from Paper 1's strict decoupling of the three properties, but the coupling is extremely weak: $\Delta K_A/K_A \sim 10^{-6}$ on galactic scales and $\sim 10^{-4}$ on cosmic scales. Lorentz invariance, the Fresnel drag coefficient, and all results derived from constant K_A hold to better than one part in 10^4 .

3. Dark Matter from $K_A(T)$

3.1 Acoustic Metric Mechanism

The orbital equation in the acoustic metric (Paper 6, Eq. 6) generalizes to position-dependent wave speed:

$$v_{\text{orbit}}^2 = \frac{GM}{r} + \frac{r}{2} \frac{dc^2}{dr} \quad (6)$$

The dark matter contribution is the second term. Using the chain rule:

$$\frac{dc^2}{dr} = \frac{1}{\rho_A} \frac{dK_A}{dr} = \frac{K_0 \eta_K}{\rho_A} \left(-\frac{dT}{dr} \right) \quad (7)$$

Since temperature *decreases* outward ($dT/dr < 0$), the wave speed *increases* outward ($dc^2/dr > 0$), producing additional centripetal acceleration—the dark matter effect with the correct sign.

3.2 Flat Rotation Curves

For a flat rotation curve ($v_{\text{orbit}} = v_{\text{flat}} = \text{const}$) at large r where the dark matter term dominates:

$$v_{\text{flat}}^2 \approx \frac{r}{2} \frac{dc^2}{dr} = \frac{K_0 \eta_K}{2 \rho_A} r \left| \frac{dT}{dr} \right| \quad (8)$$

For this to be independent of r , we require $r |dT/dr| = \text{const}$, which gives:

$$T(r) = T_{\text{center}} - \frac{2 \rho_A v_{\text{flat}}^2}{K_0 \eta_K} \ln \left(\frac{r}{r_{\text{core}}} \right) \quad (9)$$

A logarithmic temperature profile is the natural steady-state solution for viscous heating from a central gravitational flow source in a medium with constant thermal conductivity.

3.3 Quantitative Constraints

The total K_A variation across a halo (from $r_{\text{core}} = 1$ kpc to $r_{\text{halo}} = 200$ kpc, with $v_{\text{flat}} = 220$ km/s) is:

$$\frac{\Delta K_A}{K_A} = \frac{2 \rho_A v_{\text{flat}}^2}{K_A} \ln\left(\frac{r_{\text{halo}}}{r_{\text{core}}}\right) = \frac{2 \times (220 \text{ km/s})^2}{c^2} \ln(200) \approx 5.7 \times 10^{-6} \quad (10)$$

This result is *independent of η_K* : the fractional wave speed variation required for flat rotation curves depends only on v_{flat}/c and the logarithmic extent of the halo. The parameter η_K determines how this variation maps to a temperature difference:

$$\Delta T_{\text{halo}} = \frac{\Delta K_A / K_A}{\eta_K} \quad (11)$$

η_K (K^{-1})	ΔT_{halo} (K)	Comparison with He-4
10^{-5}	0.57	10× gentler than He-4 at low T
5×10^{-5}	0.11	2× gentler
1.07×10^{-4}	0.053	Same as He-4 at low T
5×10^{-4}	0.011	5× steeper

Table 1: The η_K – ΔT_{halo} degeneracy. For helium-4 below 1 K, $(1/c_1^2)|dc_1^2/dT| \sim 10^{-4} \text{ K}^{-1}$. The benchmark $\eta_K = 1.07 \times 10^{-4} \text{ K}^{-1}$ matches this value exactly.

3.4 Gravitational Lensing

Photons propagate at $c(r) = \sqrt{K_A(T(r))/\rho_A}$. The same $c(r)$ gradient that produces enhanced rotation curves also bends light through the acoustic metric. The effective mass profile is the singular isothermal sphere (SIS):

$$M_{\text{eff}}(r) = \frac{v_{\text{flat}}^2 r}{G} \quad (12)$$

and the deflection angle is:

$$\theta = \frac{4\pi \sigma^2}{c^2} \quad (13)$$

where $\sigma = v_{\text{flat}}/\sqrt{2}$. This is identical to the Paper 6 result: rotation curves and gravitational lensing are automatically consistent because both light and matter follow geodesics of the same acoustic metric.

3.5 Profile Shape and Cores

The logarithmic temperature profile (Eq. 9) produces flat rotation at large r , consistent with isothermal-sphere dark matter halos. At $r < r_{\text{core}}$, the temperature profile flattens: converging gravitational flows produce canceling velocity gradients, reducing the net shear and therefore the viscous heating rate. As $dT/dr \rightarrow 0$, the dark matter contribution vanishes, producing *naturally cored* profiles without additional assumptions—the same qualitative behavior as the first-sound mechanism of Paper 6.

Galaxy clusters. For a massive cluster with velocity dispersion $\sigma \sim 1000$ km/s, the required $\Delta K_A/K_A \approx 10^{-4}$, roughly 17 $\times$ larger than for a Milky-Way-sized galaxy. This is physically expected: clusters are more massive, with stronger gravitational inflow and correspondingly larger viscous heating—producing a larger temperature gradient and therefore a larger K_A variation.

3.6 Galactic Temperature Profile

Using the benchmark $\eta_K = 1.07 \times 10^{-4} \text{ K}^{-1}$ and anchoring the halo edge to $T_{\text{CMB}} = 2.725 \text{ K}$ (Section 6):

Location	r	$T \text{ (K)}$	$\Delta c/c$
Galaxy center	0	2.778	-2.8×10^{-6}
Core edge	1 kpc	2.778	(flat core)
Solar radius	8 kpc	2.747	-1.2×10^{-6}
Disk edge	15 kpc	2.742	-0.9×10^{-6}
Mid-halo	50 kpc	2.733	-0.4×10^{-6}
Halo edge	200 kpc	2.725	0 (reference)

Table 2: Temperature profile across a Milky-Way-sized galaxy in the hot matter model, with $v_{\text{flat}} = 220$ km/s, $\eta_K = 1.07 \times 10^{-4} \text{ K}^{-1}$, and halo edge anchored to T_{CMB} . The total variation is only 0.053 K across 200 kpc. $\Delta c/c$ is relative to the halo edge.

The entire galaxy sits within a $\sim 0.05 \text{ K}$ thermal envelope around T_{CMB} . This extraordinarily gentle gradient is sufficient for flat rotation curves because even a fractional wave speed variation of $\Delta c/c \sim 10^{-6}$ produces the full dark matter effect through the acoustic metric.

4. Dark Energy from the Cooling Cascade

4.1 K_A Gradient on Cosmic Scales

On cosmic scales, the K_A contrast between warm walls ($T_{\text{wall}} \approx 2.725$ K) and cold void centers ($T_{\text{void}} \approx 0.91$ K) produces an outward acceleration through the acoustic metric:

$$a_K = \frac{1}{2} \frac{dc^2}{dr} \approx \frac{1}{2\rho_A} \frac{\Delta K_A}{R_{\text{void}}} \quad (14)$$

The observed dark energy acceleration at the boundary of a typical void ($R_{\text{void}} \sim 50$ Mpc):

$$a_{\text{DE}} = R_{\text{void}} H_0^2 \Omega_\Lambda \approx 5.7 \times 10^{-12} \text{ m/s}^2 \quad (15)$$

Setting $a_K = a_{\text{DE}}$ gives:

$$\frac{\Delta K_{A\text{cosmic}}}{K_A} = \frac{2\rho_A R_{\text{void}} a_{\text{DE}}}{K_A} \approx 1.95 \times 10^{-4} \quad (16)$$

With $\eta_K = 1.07 \times 10^{-4} \text{ K}^{-1}$:

$$\Delta T_{\text{cosmic}} = \frac{\Delta K_{A\text{cosmic}}/K_A}{\eta_K} \approx 1.82 \text{ K} \quad (17)$$

A temperature difference of less than 2 K between walls and void centers is sufficient to produce the observed dark energy. Both the dark matter constraint ($\Delta K_A/K_A \sim 6 \times 10^{-6}$ on galactic scales) and the dark energy constraint ($\Delta K_A/K_A \sim 2 \times 10^{-4}$ on cosmic scales) are satisfied by the *same* η_K .

4.2 The Cooling Cascade

The K_A gradient is not static—it is maintained and strengthened by a self-reinforcing feedback loop:

1. **Cold void has high K_A .** The wave speed c is elevated, and the acoustic metric drives expansion of the void.
2. **Expansion cools the void.** Adiabatic cooling:

$$\frac{dT_{\text{void}}}{dt} = -\beta H T_{\text{void}} \quad (18)$$

where β parameterizes the effective adiabatic index.

3. **Cooling increases K_A .** Lower temperature $\Rightarrow$ higher $K_A \Rightarrow$ higher $c \Rightarrow$ stronger expansion drive.

4. **Return to step 1.** The cascade is self-reinforcing: each cycle of expansion produces further cooling, which increases K_A , which drives more expansion.

Meanwhile, matter-containing walls are maintained at approximately constant temperature by the viscous heating feedback (Section 2).

This cascade is *more direct* than the roton freeze-out mechanism of Paper 6(a) [8]. In Paper 6(a), the feedback goes: cooling $\rightarrow$ roton freeze-out $\rightarrow$ viscosity collapse $\rightarrow$ less braking $\rightarrow$ faster expansion (indirect, through viscosity). Here: cooling $\rightarrow K_A$ increase $\rightarrow c$ increase $\rightarrow$ stronger expansion drive (direct, through the wave speed).

The void temperature evolves as:

$$T_{\text{void}}(z) = T_{\text{void},0} (1 + z) \quad (19)$$

while $T_{\text{wall}} \approx \text{const}$. The K_A contrast at redshift z is:

$$\frac{\Delta K_A(z)}{K_A} = \eta_K [T_{\text{wall}} - T_{\text{void},0} (1 + z)] \quad (20)$$

4.3 Cascade Evolution

The temperature contrast *reverses* at:

$$z_{\text{rev}} = \frac{T_{\text{wall}}}{T_{\text{void},0}} - 1 \quad (21)$$

For $T_{\text{wall}} = 2.725$ K and $T_{\text{void},0} = 0.91$ K: $z_{\text{rev}} \approx 2.0$. At $z > 2$, voids were *warmer* than walls, and no dark energy operated.

z	T_{void} (K)	ΔT (K)	$\Delta K_A/K_A$	DE fraction
0	0.91	1.82	1.95×10^{-4}	100%
0.5	1.37	1.36	1.45×10^{-4}	74%
1.0	1.82	0.91	0.97×10^{-4}	50%
2.0	2.73	0.00	0	0%
5.0	5.46	-2.74	reversed	0%

Table 3: Evolution of the K_A contrast. At $z > 2$, voids are warmer than walls and no dark energy operates. The contrast builds gradually as voids cool, reaching full strength at $z = 0$.

This provides a natural explanation for *why dark energy is a late-universe phenomenon*: the K_A contrast had to build up through billions of years of void cooling before it became cosmologically significant.

5. The $w(z)$ Equation of State

5.1 Phantom–Quintessence Crossing

The effective dark energy density is proportional to the K_A gradient, which evolves through two competing effects:

- **Cascade growth** (strengthens dark energy): as voids cool, ΔK_A increases, adding to the expansion drive.
- **Expansion dilution** (weakens dark energy): as voids grow larger, the gradient $\Delta K_A/R_{\text{void}}$ decreases for fixed ΔK_A .

The equation of state parameter:

$$w(z) = -1 - \frac{1}{3H} \frac{d \ln \rho_{\text{DE}}}{dt} \quad (22)$$

The cascade growth rate is proportional to T_{void} (faster cooling when voids are warmer), while the expansion dilution rate is proportional to H . The crossing ($w = -1$) occurs when these rates balance:

$$T_{\text{void,cross}} = \frac{T_{\text{wall}}}{1 + \beta} \quad (23)$$

For the DESI crossing at $z_{\text{cross}} \sim 0.5$:

$$T_{\text{void},0} (1 + z_{\text{cross}}) = \frac{T_{\text{wall}}}{1 + \beta} \quad (24)$$

With $T_{\text{wall}} = 2.725 \text{ K}$ and $z_{\text{cross}} = 0.5$: $T_{\text{void},0} = T_{\text{wall}}/[1.5 (1 + \beta)]$.

- At $z > z_{\text{cross}}$: cascade growth dominates $\Rightarrow \rho_{\text{DE}}$ increasing $\Rightarrow w < -1$ (**phantom**).
- At $z < z_{\text{cross}}$: expansion dilution dominates $\Rightarrow \rho_{\text{DE}}$ decreasing $\Rightarrow w > -1$ (**quintessence**).

This is the phantom-to-quintessence crossing observed by DESI at $2.8\text{--}4.2\sigma$ significance [1].

5.2 Constraining Parameters from DESI

The crossing condition (Eq. 24) combined with the dark energy magnitude (Eq. 17) and the dark matter constraint (Eq. 10) yields a self-consistent parameter set:

Both solutions give $\eta_K \sim 10^{-4} \text{ K}^{-1}$, matching the measured temperature sensitivity of helium-4's first-sound speed at low temperature.

Parameter	$\beta = 1$	$\beta = 3$
T_{wall}	2.725 K	2.725 K
$T_{\text{void},0}$	0.91 K	0.45 K
ΔT_{cosmic}	1.82 K	2.27 K
η_K	1.07×10^{-4}	8.6×10^{-5}
ΔT_{halo}	0.053 K	0.066 K
z_{rev}	2.0	5.1

Table 4: Self-consistent parameter sets for $\beta = 1$ and $\beta = 3$, with $T_{\text{wall}} = 2.725$ K and DESI crossing at $z_{\text{cross}} = 0.5$.

5.3 Comparison with Paper 6(a)

Paper 6(a) and this paper predict the same qualitative $w(z)$ trajectory—phantom at high z , quintessence today, with a crossing near $z \sim 0.5$ –1—but through different physical mechanisms:

Feature	Paper 6(a)	This paper
Temperature near mass	Cold	Warm
Temperature in voids	Hot	Cold
Dark matter mechanism	c_1 gradient (two-fluid)	c gradient ($K_A(T)$)
Dark energy mechanism	Viscosity collapse	K_A cascade
Feedback pathway	Indirect (via ζ_n)	Direct (via $K_A \rightarrow c$)
Key parameter	$\Delta/k_B \approx 6$ K	$\eta_K \sim 10^{-4} \text{ K}^{-1}$
He-4 analog	Roton gap	Sound speed sensitivity

Table 5: Comparison of the two alternative dark sector mechanisms.

6. CMB and Local Temperature

6.1 T_{CMB} at Our Position

Paper 6 identifies the CMB as the thermal excitation spectrum of the Aether: “ $T_{\text{CMB}} = 2.725$ K is the Aether’s temperature” [7]. In the hot matter model, the Aether temperature at the Sun’s galactic position (8 kpc from center) is:

$$T_{\odot} \approx 2.747 \text{ K} \quad (25)$$

(Table 2)—only 22 mK above T_{CMB} , a 0.8% offset.

This near-equality is not a coincidence. Both T_{CMB} and T_{wall} are governed by the same cosmic expansion history: the CMB temperature tracks the relic photon field redshifted from last scattering ($T_{\text{CMB}} = T_{\text{LSS}}/(1 + z_{\text{LSS}}) \approx 2.725$ K), while the wall temperature is set by the balance of viscous

heating and expansion cooling—both of which depend on $H(t)$ and the age of the universe. The two converge to similar values at the current epoch.

The hot matter model has a structural advantage here. In the standard Paper 6 model, $T_{\text{wall}} \approx 2.7$ K but $T_{\text{void}} \approx 7$ K, giving a volume-weighted average of $0.3 \times 2.7 + 0.7 \times 7 \approx 5.7$ K—not T_{CMB} . In the hot matter model, the locally measured temperature (at our position near mass) *is* T_{CMB} to within 1%.

6.2 CMB Isotropy

The galactic temperature gradient (0.053 K across 200 kpc) would produce $\delta T/T \sim 10^{-2}$ on galactic angular scales if the CMB reflected the local Aether temperature. However, CMB photons from the last scattering surface propagate through the local Aether without thermalizing: direct photon–quasiparticle scattering is suppressed by $P_{\text{th}}/K_A \sim 10^{-38}$ [7]. The observed anisotropy ($\delta T/T \sim 10^{-5}$) reflects the primordial temperature fluctuations at $z \sim 1100$, not the current local temperature structure. The K_A gradient contributes to the integrated Sachs–Wolfe effect (photons gaining or losing energy as they traverse evolving $c(r)$ regions), at a level consistent with observations.

7. Observational Consistency

7.1 Rotation Curves

Observation: Flat rotation curves with $v_{\text{flat}} \sim 200$ km/s for Milky-Way-sized galaxies.

Model: $\Delta K_A/K_A \approx 5.7 \times 10^{-6}$ from the logarithmic temperature profile (Eq. 9).

Status: Consistent by construction.

7.2 Gravitational Lensing

Observation: Lensing mass agrees with dynamical mass from rotation curves.

Model: Both light and matter follow geodesics of the same acoustic metric with position-dependent $c(r)$. No separate mechanism needed.

Status: Automatically consistent.

7.3 Cored Dark Matter Profiles

Observation: Dwarf and low-surface-brightness galaxies show cored (not cuspy) dark matter profiles.

Model: The temperature profile flattens at $r < r_{\text{core}}$ (converging flows reduce velocity gradients), so $dT/dr \rightarrow 0$ and the dark matter contribution vanishes smoothly.

Status: Naturally cored.

7.4 Bullet Cluster

Observation: Gravitational lensing follows the galaxies, not the shocked gas, in the 1E 0657-56 cluster collision.

Model: Each galaxy's temperature profile (and therefore $K_A(r)$ gradient) is maintained by its own gravitational inflow. The shocked gas between the sub-clusters heats the local Aether, *reducing* K_A and weakening the c gradient in the collision region. The surviving K_A gradients follow the galaxies, not the gas—producing the observed lensing–gas separation.

Status: Consistent.

7.5 Hubble Tension

Observation: Local measurements give $H_0 \approx 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$; CMB-derived values give $\approx 67 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

Model: The KBC void (local underdensity, $\delta \approx 0.25$) is colder than average $\Rightarrow$ higher $K_A \Rightarrow$ higher c . Photons from distant sources traversing this region travel at higher-than-average speed, biasing local distance measurements and producing a higher apparent H_0 . The qualitative mechanism parallels Paper 6's thermal pressure excess in the void [7], but operates through K_A rather than P_{th} .

Status: Qualitatively consistent; quantitative calculation needed.

7.6 Galaxy Clusters

Observation: Clusters require $\Delta c^2/c^2 \sim 10^{-4}$, roughly $17\times$ larger than galaxies.

Model: Clusters have more massive gravitational inflows, stronger velocity gradients, and therefore larger viscous heating—producing a larger temperature contrast and K_A variation. The scaling $\Delta K_A/K_A \propto v_{\text{flat}}^2$ naturally produces the observed galaxy-to-cluster ratio.

Status: Consistent.

7.7 DESI Dark Energy Equation of State

Observation: Phantom-to-quintessence crossing at $z \gtrsim 0.5$ [1].

Model: The cooling cascade produces $w < -1$ at high z (cascade growth dominates) transitioning to $w > -1$ at low z (expansion dilution dominates), with crossing at $z \sim 0.5$ for $\beta = 1$ (Table 4).

Status: Consistent.

8. Predictions

1. $\eta_K \sim 10^{-4} \text{ K}^{-1}$. The temperature sensitivity of the Aether's bulk modulus must match the measured sensitivity of helium-4 at low temperature. If the Aether's excitation spectrum can be derived from first principles, this is a specific, testable prediction.
2. $w(z)$ **follows a cascade curve, not a polynomial.** The dark energy evolution is determined by the void cooling history (Eq. 20), not the smooth CPL parameterization. Future DESI data releases and Euclid could distinguish the two functional forms.
3. **Dark energy is absent at $z > 2$.** The temperature contrast reverses (Table 3): at $z \gtrsim 2$, voids were warmer than walls, so no K_A -driven dark energy operated. This contrasts with Λ CDM, where dark energy is present at all epochs.
4. $T_{\text{CMB}} \approx T_{\text{wall}}$. The CMB temperature directly measures the Aether temperature in mass-containing regions (where all observers reside). Any departure from $T_{\text{wall}} \approx T_{\text{CMB}}$ would falsify this model.
5. **Void regions are cold and expand faster.** Voids with lower Aether temperatures should show stronger local expansion rates and enhanced ISW imprints. This is testable with upcoming void catalogs from DESI and Euclid.
6. **Single parameter determines both dark phenomena.** The same η_K that produces dark matter on galactic scales produces dark energy on cosmic scales. Independent measurements of both should converge on the same value—a powerful consistency test.

9. Open Questions

1. **Derivation of η_K from Aether microscopic parameters.** Can $\eta_K \sim 10^{-4} \text{ K}^{-1}$ be derived from m_A , ρ_A , and K_A ? In helium-4, the temperature dependence of the sound speed arises from the phonon-roton excitation spectrum; the analogous calculation for the Aether requires knowledge of the full excitation spectrum.

2. **Three-dimensional temperature profile.** The logarithmic $T(r)$ profile (Eq. 9) produces flat rotation, but the full three-dimensional profile—including the transition from halo to void—needs to be derived from the viscous heating equation with appropriate boundary conditions.
3. **Tully–Fisher relation.** The model predicts $v_{\text{flat}}^2 \propto (\text{heating rate}) \times \ln(r_{\text{halo}}/r_{\text{core}})$. Whether this reproduces the observed $M \propto v^4$ scaling depends on how the viscous heating rate scales with baryonic mass. For $\dot{\epsilon} \propto M^{0.7}$ (sublinear, from the star-formation main sequence), the predicted slope approaches the observed value.
4. **CMB power spectrum.** How does the $K_A(T)$ mechanism affect baryon-photon acoustic oscillations at $z \sim 1100$? At the CMB epoch, the temperature was nearly uniform ($\delta T/T \sim 10^{-5}$), so $\delta K_A/K_A \sim \eta_K \times \delta T \sim 10^{-4} \times 10^{-5} \times 3000 \sim 3 \times 10^{-6}$ —small but potentially detectable in the damping tail.
5. **Reconciliation with Paper 6.** The standard model (cold near mass) and the hot matter model (warm near mass) produce qualitatively similar observational signatures through different mechanisms. Are they limiting cases of a more general framework? Could the truth lie between the two extremes?
6. **The β parameter.** The adiabatic cooling rate of void Aether determines the cascade evolution and the $w(z)$ crossing redshift. In a standard fluid, $\beta = 3$; for the nearly incompressible Aether, β depends on how thermal energy couples to the expansion and requires a first-principles calculation.
7. **Specific heat.** The viscous heating rate constrains the product $\rho_A c_v T_{\text{wall}}$. For $T_{\text{wall}} \approx 2.7$ K, the required specific heat is $c_v \sim 5 \times 10^{-20}$ J/(kg K)—consistent with a superfluid far below T_c (Debye $c_v \propto (T/T_c)^3$), but the consistency with the same excitation spectrum that determines η_K needs to be verified.
8. **Hubble tension magnitude.** The qualitative mechanism (cold KBC void $\Rightarrow$ higher $K_A \Rightarrow$ higher local c) is clear, but the quantitative requirement ($\sim 8\%$ discrepancy in H_0) needs detailed modeling of the KBC void temperature profile and its effect on distance-ladder measurements.

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